इंटरनेट मानक

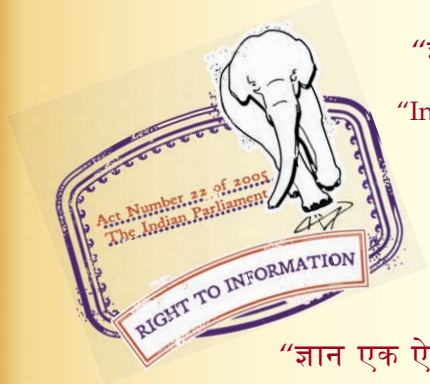
Disclosure to Promote the Right To Information

Whereas the Parliament of India has set out to provide a practical regime of right to information for citizens to secure access to information under the control of public authorities, in order to promote transparency and accountability in the working of every public authority, and whereas the attached publication of the Bureau of Indian Standards is of particular interest to the public, particularly disadvantaged communities and those engaged in the pursuit of education and knowledge, the attached public safety standard is made available to promote the timely dissemination of this information in an accurate manner to the public.

"जानने का अधिकार, जीने का अधिकार"
Mazdoor Kisan Shakti Sangathan
"The Right to Information, The Right to Live"

"पुराने को छोड नये के तरफ"
Jawaharlal Nehru
"Step Out From the Old to the New"

IS 9550 (2001): Bright Steel Bars [MTD 4: Wrought Steel Products]



"ज्ञान से एक नये भारत का निर्माण"
Satyanarayan Gangaram Pitroda
"Invent a New India Using Knowledge"



"ज्ञान एक ऐसा खजाना है जो कभी चुराया नहीं जा सकता है"
Bhartṛhari—Nītiśatakam
"Knowledge is such a treasure which cannot be stolen"

BLANK PAGE





PROTECTED BY COPYRIGHT

भारतीय मानक

चमकीले इस्पात की छड़ें — विशिष्टि

(पहला पुनरीक्षण)

Indian Standard

BRIGHT STEEL BARS — SPECIFICATION

(First Revision)

ICS 77.140.60

© BIS 2001

BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

March 2001

Price Group 4

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Wrought Steel Products Sectional Committee had been approved by the Metallurgical Engineering Division Council.

This standard was first published in 1980. While reviewing this standard, in the light of experience gained during these years, the committee decided to revise it to align with the present practices being followed by the Indian industry.

In this revision, the following modifications have been made:

a) Scope of the standard has been modified.
b) Reference clause has been included.
c) Dimensional tolerances have been modified.
d) Surface conditions have been modified.

In the preparation of this standard, assistance has been derived from the following overseas standards:

a) prEN 10277-1 Bright steel products — Technical delivery condition : Part 1 General (October 1996)
b) prEN 10278 Dimensions and tolerances of bright steel products (September 1996)

An informative Annex A has been given for the benefit of purchaser giving particulars to be specified by the purchaser while placing order for the bright bars covered in this standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

BRIGHT STEEL BARS — SPECIFICATION

(First Revision)

1 SCOPE

This standard covers cold drawn, turned, turned and reeled, or ground steel bars of grades and cross sections regarded as generally suitable for heat treatment, for machining into component or for use in 'as finished' condition in constructional applications or other similar purposes, such as for the manufacture of threaded and machined components for general engineering purposes.

2 REFERENCES

The Indian Standards listed in Annex B, contain provisions which through reference in this text, constitute provision of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated in Annex B.

3 TERMINOLOGY

3.1 For the purpose of this standard the definitions given in IS 1956 (Part 5) and the following shall apply.

3.2 Drawn Products

Products of various cross section shapes obtained, after descaling, by drawing of hot rolled bars or rods on a draw bench (cold deformation without removing material). This operation gives the product special features with respect to shape, dimensional accuracy, surface finish and mechanical properties. In addition, the process causes cold working of the product, which can be eliminated by subsequent heat treatment. Product in lengths are delivered straightened regardless of size.

3.3 Turned Products

Round bars produced by turning on a lathe or a centreless turning machine after straightening, which may be followed by polishing. This operation gives the bar special features with respect to shape, dimensional accuracy and surface finish. The removal of metal is carried out in such a way that the bright product is generally free from rolling defects and surface decarburization.

3.4 Ground Products

Drawn or turned round bars given an improved surface quality and dimensional accuracy by grinding and polishing.

4 SUPPLY OF MATERIAL

4.1 General requirements relating to the supply of material shall be as laid down in IS 8910.

4.2 The steel products shall be delivered in or a combination of the following finished conditions:

a) Drawn,
b) Turned,
c) Ground, and
d) Turned and reeled.

4.3 Unless otherwise agreed, bars up to 45 mm in dimensions are normally delivered with sheared ends. The ends of the product shall be as specified by the purchaser at the time of enquiry and order, for example, chamfering, facing, etc.

5 MANUFACTURE

5.1 The processes used in making the steel and in manufacturing bright steel products are left to the discretion of the manufacturer.

5.2 Unless specified otherwise, the steel shall be supplied in the rimmed, semi-killed or killed condition, as per mutual agreement between the purchaser and the manufacturer.

6 CHEMICAL COMPOSITION

The material specification and heat treatment, if any, shall be so selected that the rolled bars used for the manufacture of bright bars will ensure the desired mechanical properties as per the requirements of the purchaser.

7 MECHANICAL PROPERTIES

The mechanical properties of bright bars shall be as agreed to between the purchaser and the manufacturer.

8 DIMENSIONS AND TOLERANCES

8.1 Dimensions

Bars, sections and flats shall be supplied as per the dimensions specified in the orders.

8.2 Tolerances

8.2.1 *Diameter, Thickness and Width*

Tolerances on dimensions shall be as specified by the purchaser and shall be in accordance with IS 919 (Part 2) as given in Table 1.

Table 1 Tolerance Class According to Finished Conditions

(*Clauses* 8.2.1 *and* 8.2.1.1)

Finished Condition	Tolerance Class to IS 919 (Part 2)						
	h6	h7	h8	h9	h10	h11	h12
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Drawn				R	R	R, S, H	R, S, H
Turned				R	R	R	R
Turned and Reeled				R	R	R	R
Ground	R	R	R	R	R	R	R

R = round, S = square, H = hexagon

8.2.1.1 Unless specified otherwise, tolerances on dimensions shall be as follows:

a) For drawn round bars other than those under (e), or turned bars: h10 to Table 2.

b) For hexagonal and square drawn bars: h11 for dimensions up to and including 80 mm, h12 for dimensions over 80 mm according to Table 1 and 2.

c) For drawn flats: in accordance with Table 3 and 4;

d) For ground products: in accordance with Table 1 and 2.

e) For drawn round bars in the final quenched and tempered condition: h11.

8.2.2 *Length*

Bars and sections shall be supplied in length of 2.5 m to 4.5 m with maximum of 10 percent shorts of not less than 1.5 m. If bars of other lengths are required to be supplied the same shall be as per the mutual agreement between the purchaser and the manufacturer.

8.2.2.1 When bars are required in specific lengths, tolerance of $^{+5}_{0}$ mm shall be applicable.

8.2.2.2 The ends of the bars shall be cut square without disturbing the dimensional tolerances.

Table 2 Tolerance Classes

(*Clause* 8.2.1.1)

Nominal Dimensions		Tolerance Class to IS 919 (Part 2)						
Over	Up to and Including	h6	h7	h8	h9	h10	h11	h12
mm	mm							
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
6	10	0.009	0.015	0.022	0.036	0.058	0.090	0.150
10	18	0.011	0.018	0.027	0.043	0.070	0.110	0.180
18	30	0.013	0.021	0.033	0.052	0.084	0.130	0.210
30	50	0.016	0.025	0.039	0.062	0.100	0.160	0.250
50	80	0.019	0.030	0.046	0.074	0.120	0.190	0.300
80	100	0.022	0.035	0.054	0.087	0.140	0.220	0.350

NOTE — The above deviation values are negatively disposed about the nominal dimension. For example a 20 mm nominal diameter having a tolerance class h9 is 20 mm +0, – 0.052 or 19.948/20.000 mm.

Table 3 Width Tolerance for Drawn Flats

(*Clause* 8.2.1.1)

Width		Deviation		IS 919 (Part 2) Tolerance Class to
Over	Up to and Including	Plus	Minus	
mm	mm	mm	mm	
(1)	(2)	(3)	(4	(5)
—	18	0	0.11	h11
18	30	0	0.13	h11
30	50	0	0.16	h11
50	80	0	0.19	h11
80	100	0	0.22	h11
100	150	0.50	0.50	—
150	200	1.00	1.00	—
200	300	2.00	2.00	—
300	400	2.50	2.50	—

Table 4 Thickness Tolerance for Drawn Flats

(*Clause* 8.2.1.1)

Thickness		Deviation for Widths		IS 919 (Part 2) Tolerance Class to
Over	Up to and Including	Up to and Including 65	Over 65	
mm	mm	mm	mm	
(1)	(2)	(3)	(4)	(5)
3	6	– 0.075	—	—
6	10	– 0.090	– 0.11	h11
10	18	– 0.11	—	—
18	30	– 0.13	– 0.13	h11
30	50	– 0.16	– 0.16	h11
50	60	– 0.19	– 0.19	h11
60	80	– 0.30	– 0.30	h12
80	100	– 0.35	– 0.35	h12

NOTE — The above deviation values are negatively disposed.

8.2.3 *Straightness Tolerance*

8.2.3.1 Unless otherwise agreed, the permissible deviations shall not exceed 1.5 mm in any 1 m length. The methods for evaluating straightness are given in Annex C for guidance.

8.2.3.2 Any other details regarding measuring and sampling method for straightness tolerance of bright bars shall be agreed upon at the time of inquiry and order.

8.2.4 *Out of Shape*

Maximum deviation from 'out of shape' shall be not more than half the specified tolerance.

8.2.5 *Edges of Non-round Bars*

Non-round bars, that is, square, hexagon and flat in widths up to and including 150 mm shall have sharp corners without radius. For widths over 150 mm the corner profile may be undefined within a distance of 0.5 mm of the hypothetical edge, unless sharp corners

have specifically been ordered.

9 SURFACE CONDITION

The surface quality of the steel product shall be one of the classes according to Table 5. Drawn products shall have a smooth, scale free surface. Products in the final heat treated condition shall be free from loose surface scale, but may have surface discoloration or darkening.

NOTES

1 Drawn products may have minor surface imperfections, for example, pores, pits, scoring.

2 Non-circular sections will not have the same quality of surface finish as round sections.

3 Products in drawn and turned condition are not supplied with a specified surface finish.

4 Longitudinal surface cracks cannot be entirely eliminated from surfaces of drawn products without removal of material.

5 Products in the 'technically crack free by testing' condition are only available in the turned or turned and ground condition.

6 Bars shall be reasonably free from harmful internal defects.

9.1 Decarburization

9.1.1 No surface decarburization shall be permitted for turned and ground bars.

9.1.2 In cold drawn bars, total decarburization shall not be permitted. The maximum extent of partial decarburization that can be permitted for cold drawn bars will be guided by the maximum depth of defects for Class 2 bars as given in Table 5. If, however, a lower depth of decarburization is required, it shall be mutually agreed to between the purchaser and the manufacturer.

10 SAMPLING

10.1 Sampling for Chemical Analysis

If the product analysis is required by the purchaser, at least one sample product shall be taken from each cast/lot.

10.1.1 For product analysis, the selection of sample shall be carried as per mutual agreement between the purchaser and the supplier. Product analysis shall not be applicable for rimming steel.

10.2 Sampling for Mechanical Tests

For the purpose of mechanical tests, samples shall be selected on the following basis:

Condition	*Number of Samples*
Turned or ground	One for every 20 tonnes or part thereof with a minimum one per cast
Cold drawn	One for every cast
Annealed and processed	One for every batch (not over 20 tonnes) with a minimum of one per cast

10.2.1 Test pieces for mechanical properties shall be taken in the direction of the fibre, the rolling direction.

10.3 Selection and preparation of samples and test pieces shall be done in accordance with IS 3711.

11 OTHER TESTS

11.1 If required and mutually agreed to between the purchaser and the manufacturer, the following tests may be carried out and test certificates furnished by the manufacturer:

a) Hardenability (*see* IS 3848);

b) Inclusion content (*see* IS 4163);

Table 5 Surface Quality Classes

(*Clause* 9)

Dimension, *d*, mm		Permissible Defect Depth, *Max* mm			
Over	Up to and Including	Class 1	Class 2	Class 3	Class 4
(1)	(2)	(3)	(4)	(5)	(6)
—	15	0.03	0.03	0.02	Technically crack free by testing
15	20	0.02*d*	0.02*d*	0.02	
20	75	0.02*d*	0.02*d*	0.01*d*	
75	—	0.11	0.50	0.75	

NOTE — *d* is the nominal diameter of bar and distance across flats of square and hexagons.

c) Grain size (*see* IS 4748);

d) Microstructure [*see* IS 7739 (Part 1), IS 7739 (Part 2) and IS 7739 (Part 5)];

e) Crack testing (*see* IS 2595, IS 3658, IS 3664 and IS 3703);

f) Decarburization (*see* IS 6396);

g) Microscopic examination for depth of defects; and

h) Any other tests (that is, surface roughness in micron), etc.

11.1.1 Acceptance values and details of tests in absence of any Indian Standard shall be as agreed to between the purchaser and the manufacturer.

12 RETEST

Should any one of the test pieces first selected fail to pass any of the tests specified in this standard, two further samples shall be selected from the same lot for testing in respect of each failure. Should the test pieces from both these additional samples pass, the material represented by the test samples shall be deemed to comply with the requirement of that particular test. Should the test pieces from either of these additional samples fail, the material represented by the test samples shall be deemed as not conforming to this standard.

13 SURFACE PROTECTION AND PACKING

A suitable clear temporary rust preventive shall be applied on all the bars to avoid rust during transit (*see* IS 1153 and IS 1154). The material shall be suitably packed in bundles - hessian wrapped to prevent sagging, corrosion and damage during transit.

14 MARKING

14.1 Each bar over 50 mm in diameter or of equivalent cross sectional area shall be stamped at one end with the cast identification and supplier's identification mark/ code. Bars of 50 mm in diameter or of equivalent cross sectional area and below shall be bundled together and tied with suitable steel strappings at 3-4 places along the length of the bars. The metal tag shall be securely attached to each bundle and shall bear the information such as supplier's name, cast/heat number, size and mass, order No., etc.

14.1.1 The material may also be marked with the Standard Mark.

14.1.2 The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act*, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

ANNEX A

(Foreword)

BASIS FOR ORDER

A-1 While placing an order for the steels covered by this standard, the purchaser should specify clearly the following:

a) Chemical composition;
b) Surface finish (if it is felt necessary by the purchaser, the quantitative value in microns may be specified);
c) Mechanical properties;
d) Dimensional tolerance;
e) Surface defects;
f) Heat treatment:
   1) Annealed and cold finished,
   2) Normalized and cold finished,
   3) Cold drawn and stress relieved,
   4) Carbon restoration annealed to overcome surface decarburization on cold drawn bars (for round bars produced by machining, surface decarburization is removed during the manufacturing process), and
g) Condition :
   1) Cold drawn,
   2) Cold drawn and machined,
   3) Cold drawn and ground,
   4) Hot wrought and machined, and
   5) Hot wrought machined and ground.

A-2 The following delivery conditions have been given for guidance:

a) *Round Bars*

Up to and including 50 mm diameter	—	Cold drawn
Above 50 mm diameter	—	Cold drawn/hot rolled and turned.

b) *Rectangular/Square/Hexagonal Bars*

All sizes — Cold drawn

c) *Sections*

All sizes — Cold drawn

ANNEX B

(*Clause* 2)

LIST OF REFERRED INDIAN STANDARDS

IS No.	*Title*
919 (Part 2) : 1993	ISO system of limits and fits : Part 2 Tables of standard tolerance grades and limit deviations for holes and shafts (*first revision*)
1153 : 1975	Temporary corrosion preventive, fluid, hard film, solvent deposited (*first revision*)
1154 : 1957	Temporary corrosion preventive, fluid, soft film, solvent deposited, water displacing
1956 (Part 5) : 1976	Glossary of terms relating to iron and steel : Part 5 Bright steel bar and steel wire (*first revision*)
2595 : 1978	Code of practice for radiographic testing (*first revision*)
3658 : 1999	Code of practice for liquid penetrant flaw detection (*second revision*)
3664 : 1981	Code of practice for ultrasonic pulse echo testings by contact and immersion methods (*first revision*)
3703 : 1980	Code of practice for magnetic particle flaw detection (*first revision*)
3711 : 1990	Wrought steel-selection and preparation of samples and test pieces for mechanical tests (*first revision*)
3848 : 1981	Method for end quench test for hardenability of steel (*first revision*)
4163 : 1982	Method for determination of inclusion content in steel by microscopic method (*first revision*)
4748 : 1988	Methods of estimating average grain size of metals (*first revision*)
6396 : 1983	Method for measuring decarburized depth of steel (*first revision*)
7739	Code of practice for preparation of metallurgical specimens:
(Part 1) : 1975	General features
(Part 2) : 1975	Electrolytic polishing
(Part 5) : 1976	Iron and steel and their examination
8910 : 1978	General technical delivery requirements for steel and steel products

ANNEX C

(*Clause* 8.2.3.1)

METHODS FOR EVALUATING STRAIGHTNESS

C-1 SCOPE

This annex specifies two methods for the evaluation of the straightness of the bright steel bars as provided for in **8.2.3**. The method specified in **C-2** is the preferred method and **C-3** is an alternative method. The choice of method shall be as agreed at the time of enquiry and order.

C-2 PREFERRED METHOD

C-2.1 The bar shall be supported on a suitably so as to eliminate or minimize sagging.

C-2.2 A 1-m long straight edge shall be placed on the surface of the bar at any position along its length. No part of the straight edge shall be within 150 mm of the ends of the bar.

C-2.3 Straightness shall be determined by measuring the maximum gap between the bar and the straight edge by suitable means, for example, feeler gauge. The bar shall be deemed straight where the maximum gap does not exceed the values specified in **8.2.3.1**.

C-3 ALTERNATIVE METHOD FOR ROUND BARS

C-3.1 The round bar shall be supported on centres placed 1 m apart and capable of being rotated.

C-3.2 Straightness shall be measured by means of a suitable dial or indicator gauge placed at any position between the supporting centres.

C-3.3 The bar shall be deemed straight when rotating the bar through 360° the indicated reading is not greater than twice the deviation specified in **8.2.3.1**.

Bureau of Indian Standards

BIS is a statutory institution established under the *Bureau of Indian Standards Act,* 1986 to promote harmonious development of the activities of standardization, marking and quality certification of goods and attending to connected matters in the country.

Copyright

BIS has the copyright of all its publications. No part of these publications may be reproduced in any form without the prior permission in writing of BIS. This does not preclude the free use, in the course of implementing the standard, of necessary details, such as symbols and sizes, type or grade designations. Enquiries relating to copyright be addressed to the Director (Publications), BIS.

Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards : Monthly Additions'.

This Indian Standard has been developed from Doc : No. MTD 4 (4211).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110 002
Telephones : 323 01 31, 323 33 75, 323 94 02

Telegrams: Manaksanstha
(Common to all offices)

Regional Offices :	Telephone
Central : Manak Bhavan, 9 Bahadur Shah Zafar Marg NEW DELHI 110 002	323 76 17 323 38 41
Eastern : 1/14 C. I. T. Scheme VII M, V. I. P. Road, Kankurgachi CALCUTTA 700 054	337 84 99, 337 85 61 337 86 26, 337 91 20
Northern : SCO 335-336, Sector 34-A, CHANDIGARH 160 022	60 38 43 60 20 25
Southern : C. I. T. Campus, IV Cross Road, CHENNAI 600 113	235 02 16, 235 04 42 235 15 19, 235 23 15
Western : Manakalaya, E9 MIDC, Marol, Andheri (East) MUMBAI 400 093	832 92 95, 832 78 58 832 78 91, 832 78 92

Branches : AHMADABAD. BANGALORE. BHOPAL. BHUBANESHWAR. COIMBATORE. FARIDABAD. GHAZIABAD. GUWAHATI. HYDERABAD. JAIPUR. KANPUR. LUCKNOW. NAGPUR. PATNA. PUNE. RAJKOT. THIRUVANANTHAPURAM.

Printed at New India Printing Press, Khurja, India